

Assignment-2

Section 1. Multiple Choices. (For each question, please select **the most appropriate answer.**)

1.1. Which of the following complexes will be most reactive towards oxidative addition of H_2 ?

- (a) $RhCl(CO)(PPh_3)_2$ (b) $RhBr(CO)(PPh_3)_2$
(c) $IrCl(CO)(PPh_3)_2$ (d) $IrBr(CO)(PPh_3)_2$

d

1.2. Which of the following oxidative addition reactions is/are not likely through a concerted mechanism?

- (a) $IrI(CO)(PPh_3)_2 + Me_3SiH \longrightarrow IrI(H)(SiMe_3)(CO)(PPh_3)_2$
(b) $IrI(CO)(PPh_3)_2 + MeCOCl \longrightarrow IrI(Cl)(MeCO)(CO)(PPh_3)_2$
(c) $IrI(CO)(PPh_3)_2 + H_2 \longrightarrow IrI(H)_2(CO)(PPh_3)_2$
(d) $IrCl(CO)(PPh_3)_2 + HCl$ (in aqueous solution) $\longrightarrow IrHCl_2(CO)(PPh_3)_2$
(e) two of them (f) three of them (g) all of them (h) none of them

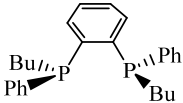
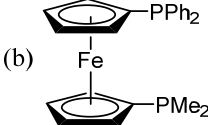
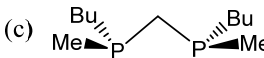
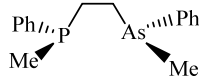
(b d) -- (e)

1.3. Which of the following reactions would you expect to have the largest equilibrium constant K_{eq} ?

- (a) $Cp(CO)_3Cr-CH_3 + CO \rightleftharpoons Cp(CO)_3Cr-COCH_3$
(b) $Cp(CO)_3Mo-CH_3 + CO \rightleftharpoons Cp(CO)_3Mo-COCH_3$
(c) $Cp(CO)_3W-CH_3 + CO \rightleftharpoons Cp(CO)_3W-COCH_3$
(d) $Cp(CO)_3W-H + CO \rightleftharpoons Cp(CO)_3W-COH$

a

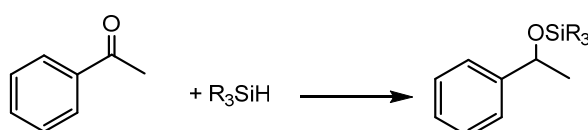
1.4. Which of the following ligands could be used for asymmetric catalytic hydrogenation of $C_6H_5CH=CHMeCONMe_2$?

- (a)  (b)  (c) 
(d) 

a, d -- (e)

- (e) two of them (f) three of them (g) all of them (h) none of them

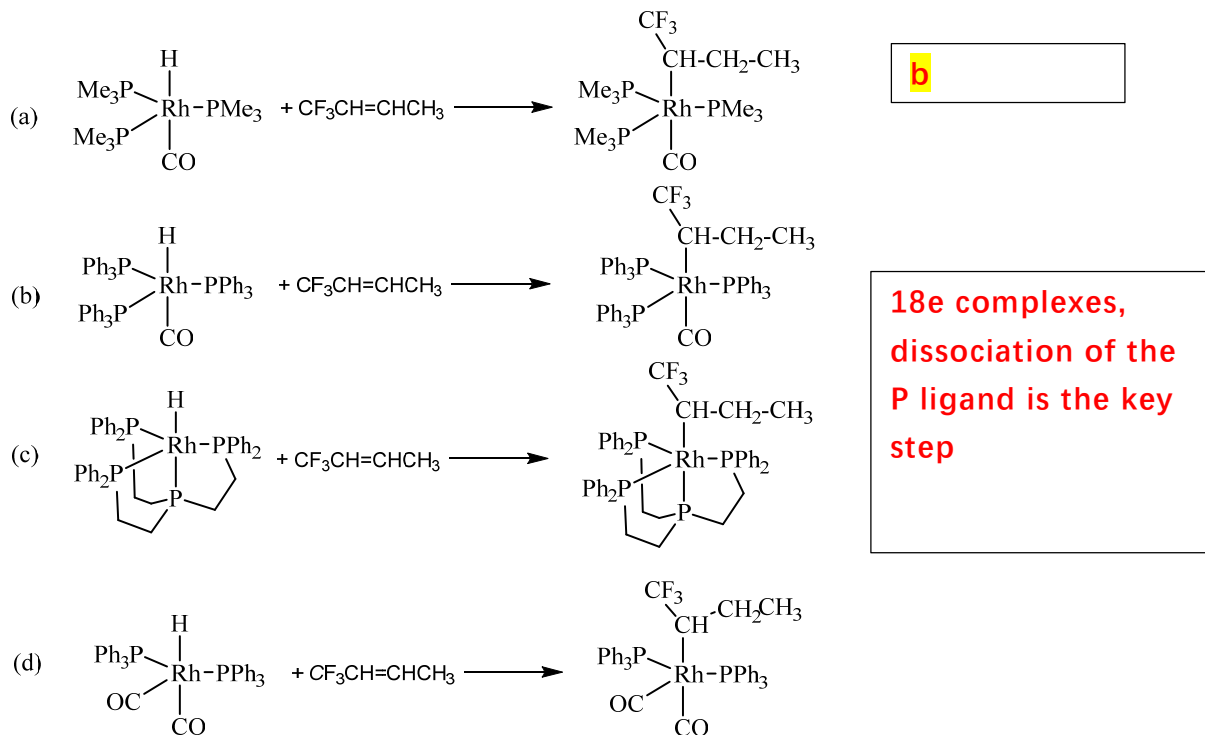
1.5. Which of the following complexes could be used as the catalyst or catalytic precursor for the following reaction?



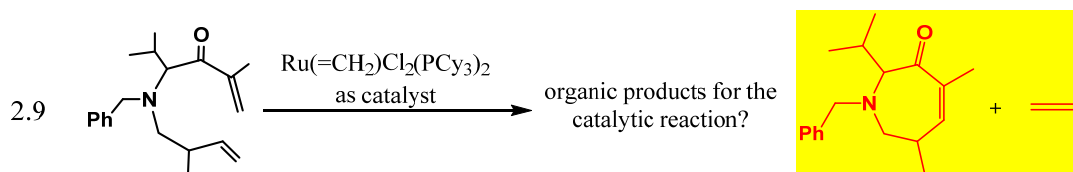
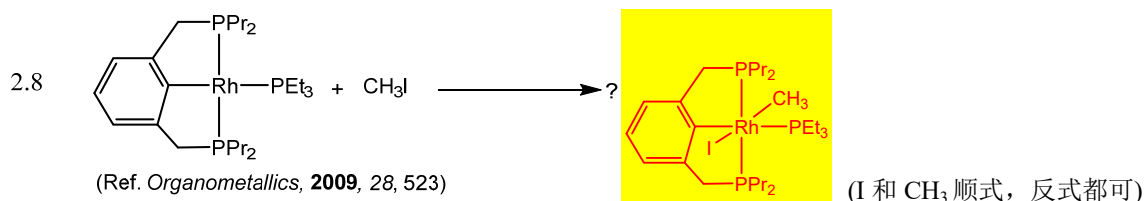
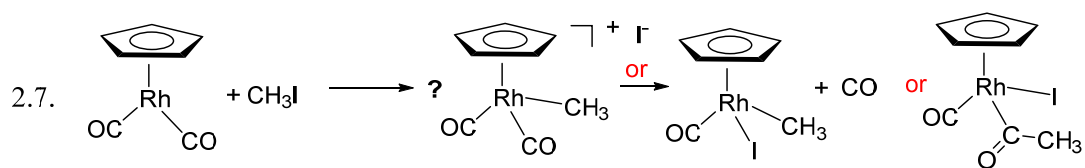
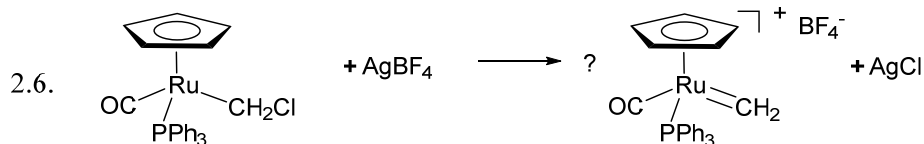
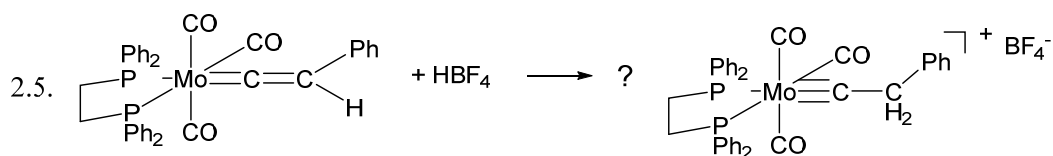
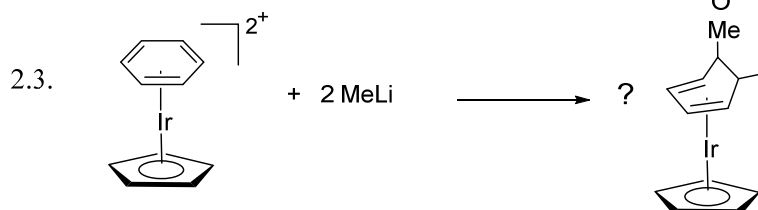
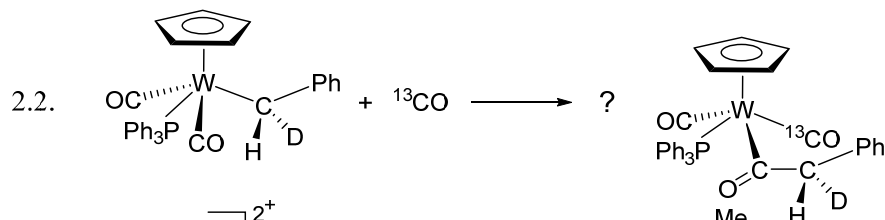
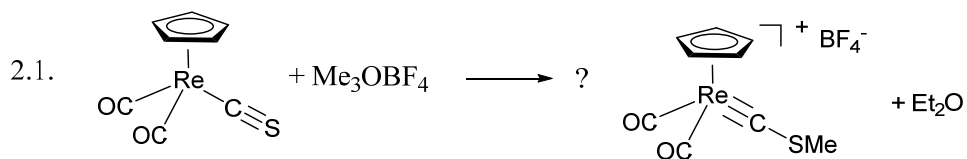
a, b, c — (f)

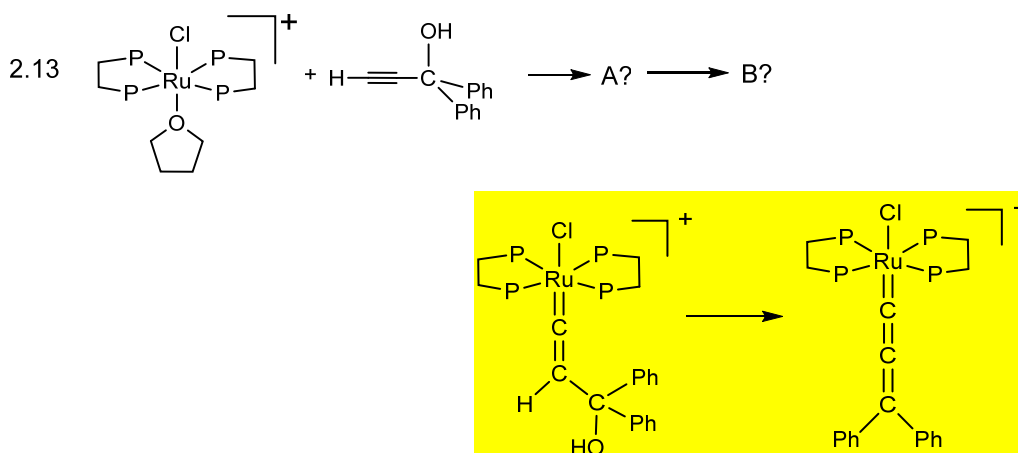
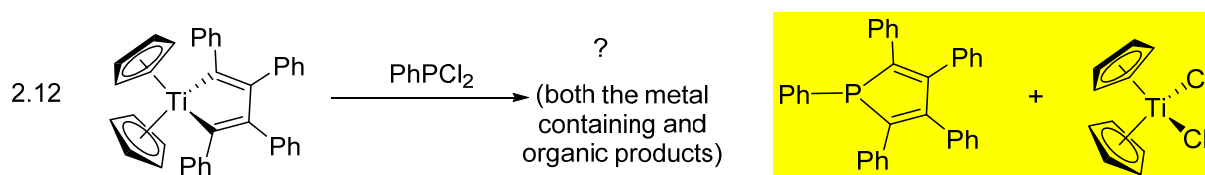
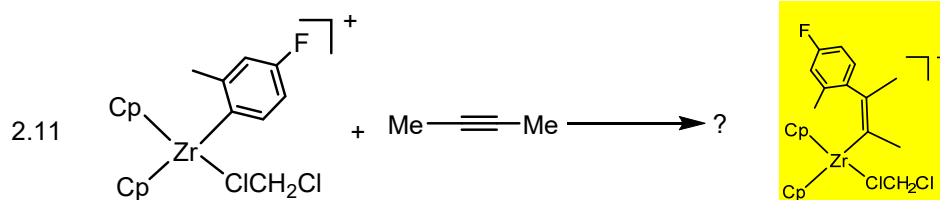
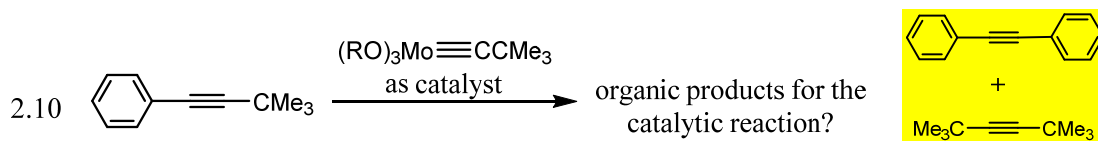
- (a) $\text{Pt}(\text{PPh}_3)_4$ (b) $\text{RhCl}(\text{PPh}_3)_3$ (c) $\text{PtHCl}(\text{PPh}_3)_2$ (d) SiH_4
 (e) two of them (f) three of them (g) all of them (h) none of them

1.6. Which of the following reactions would you expect to have the highest reaction rate?

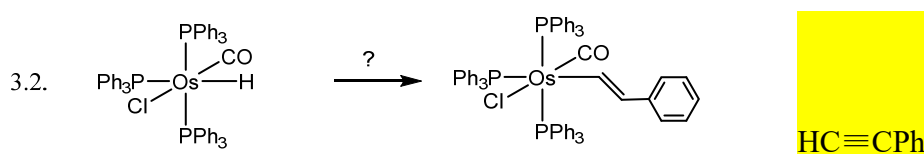
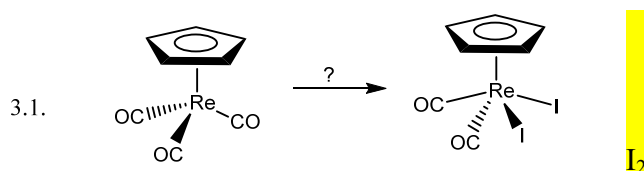


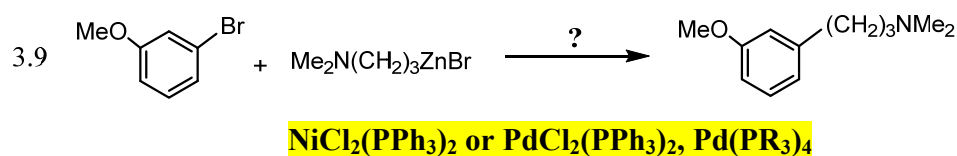
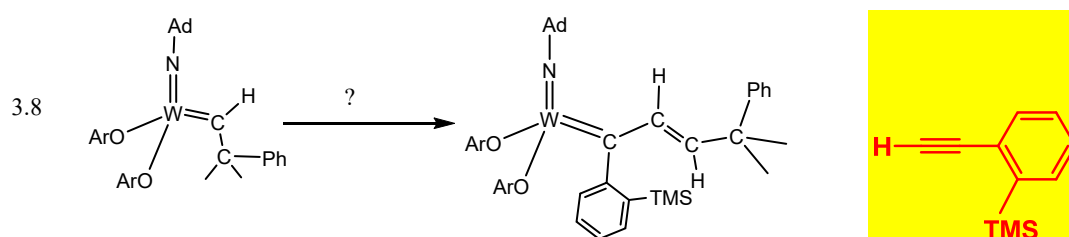
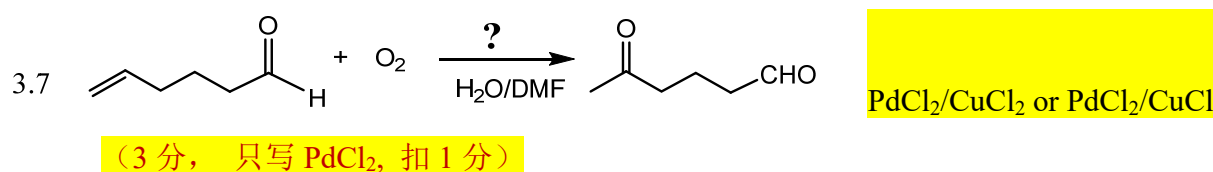
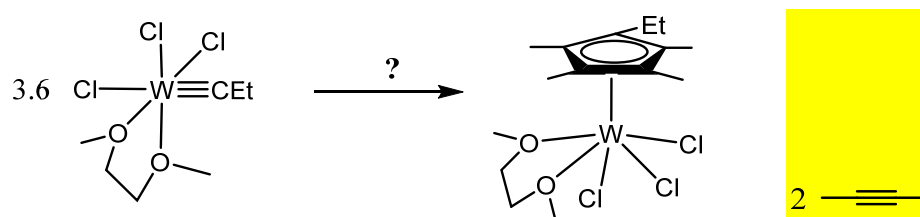
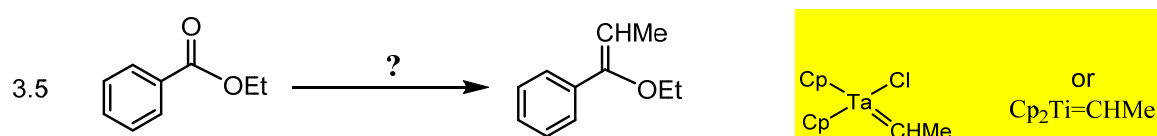
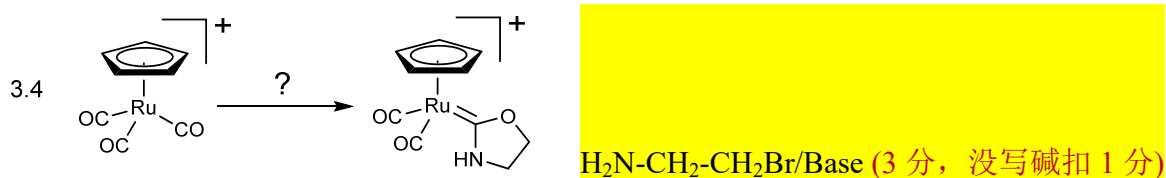
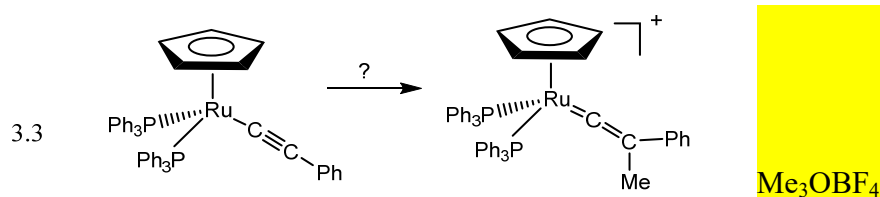
Section 2. Predict the products of the following reactions (30 points total)



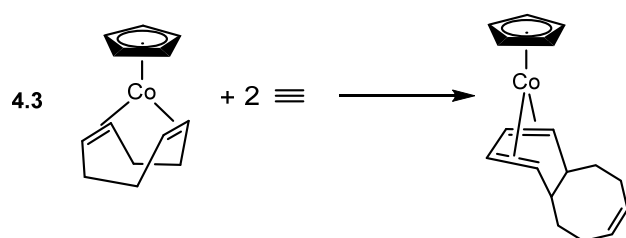
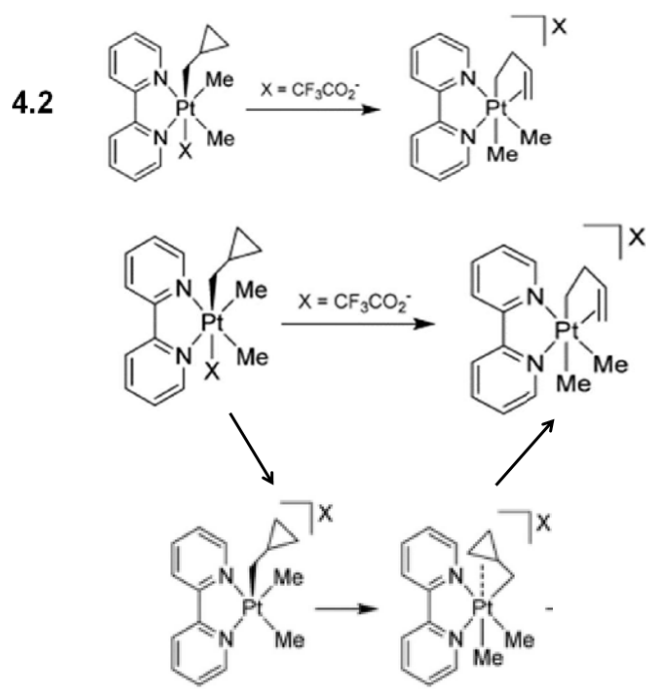
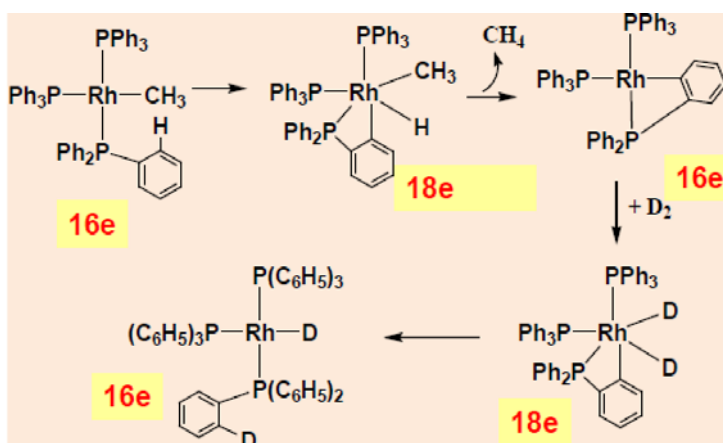
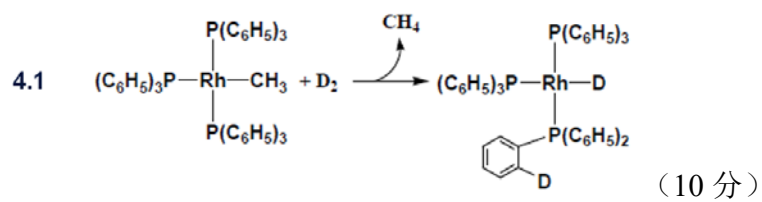


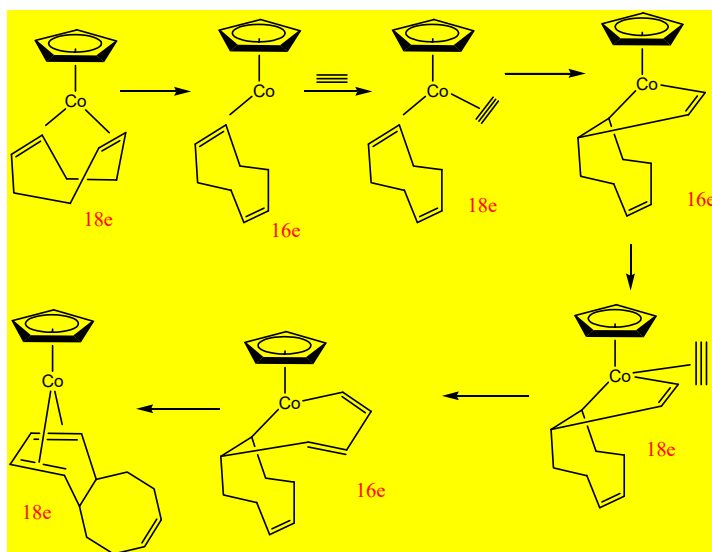
Section 3. Provide reagents for the following reactions (Note: Some of the reactions may need more than one reagents and/or catalyst(s) (20 Points)).





Section 4. Propose reasonable mechanisms for stoichiometric reactions (60 points).



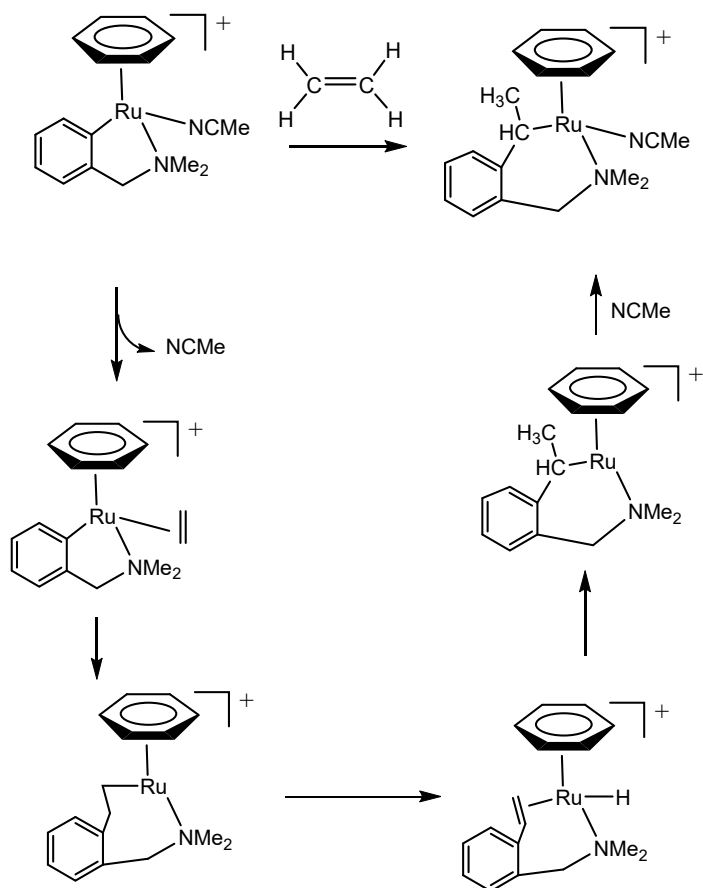


η^2 -COD, 1;
 π -Alkyne 1
 Oxidative Coupling, 3;
 π -Alkyne 1
 Insertion 2
 RE, 2;

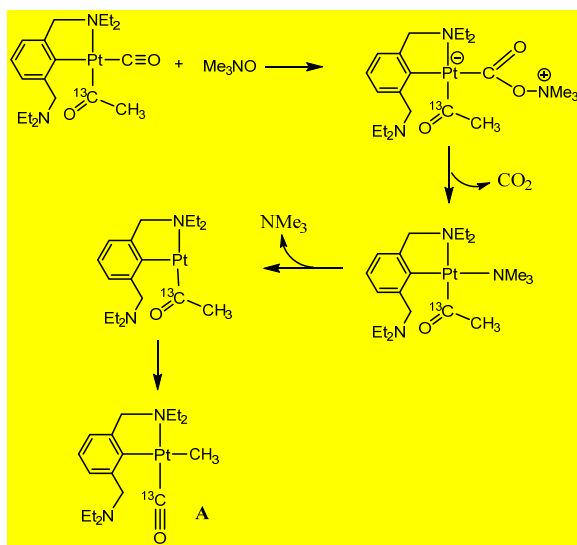
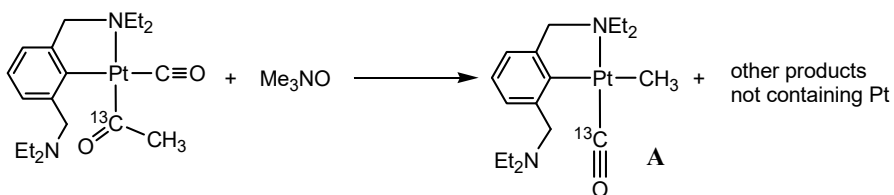
Or: (Coupling of two alkynes first)

COD Dissociation, 1
 π -coordination of two alkyne, 1
 Oxidative Coupling, 3;
 η^2 -COD, 1
 Insertion, 2
 RE, 2;

4.4

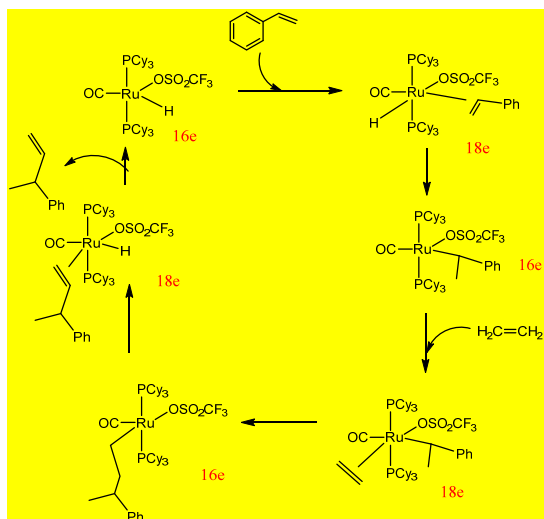
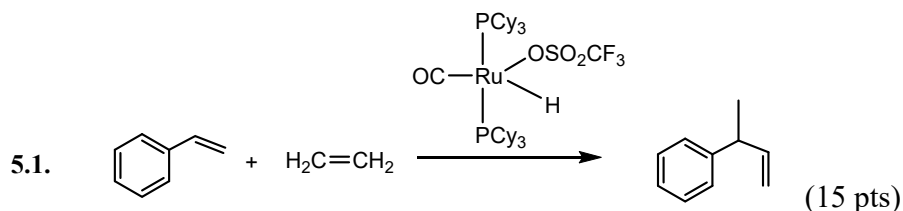


4.5 The following reaction was reported recently (see *Organometallics*, 2009, **28**, 1613). Propose a mechanism for the formation of product A. Be sure to indicate what other products are generated in the reaction.

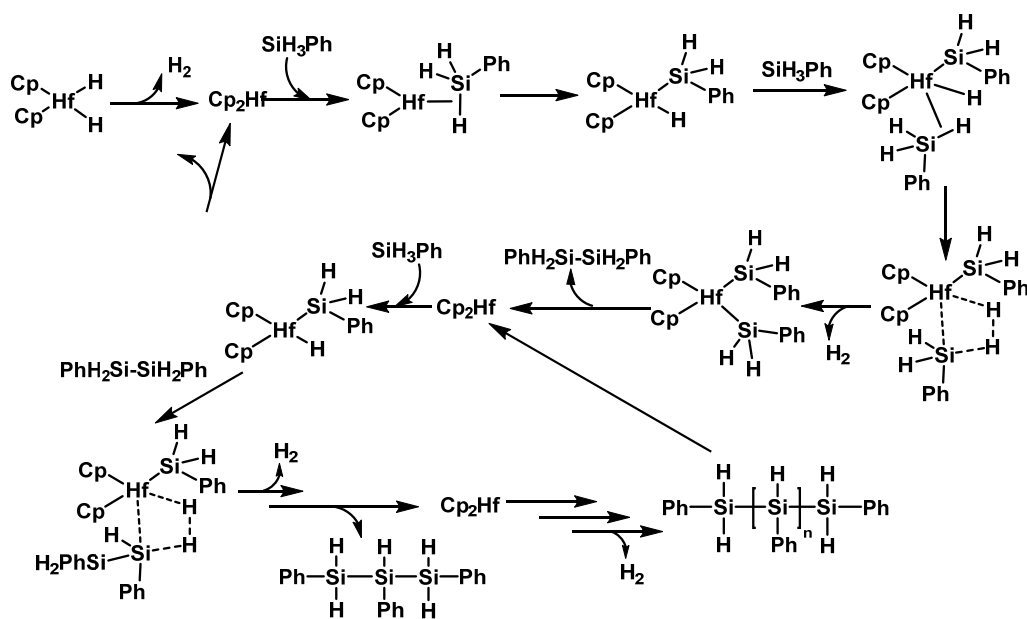
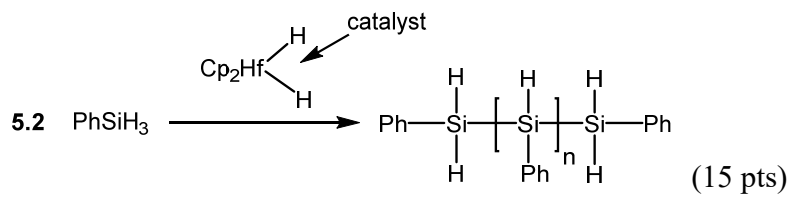


2 pts each step.

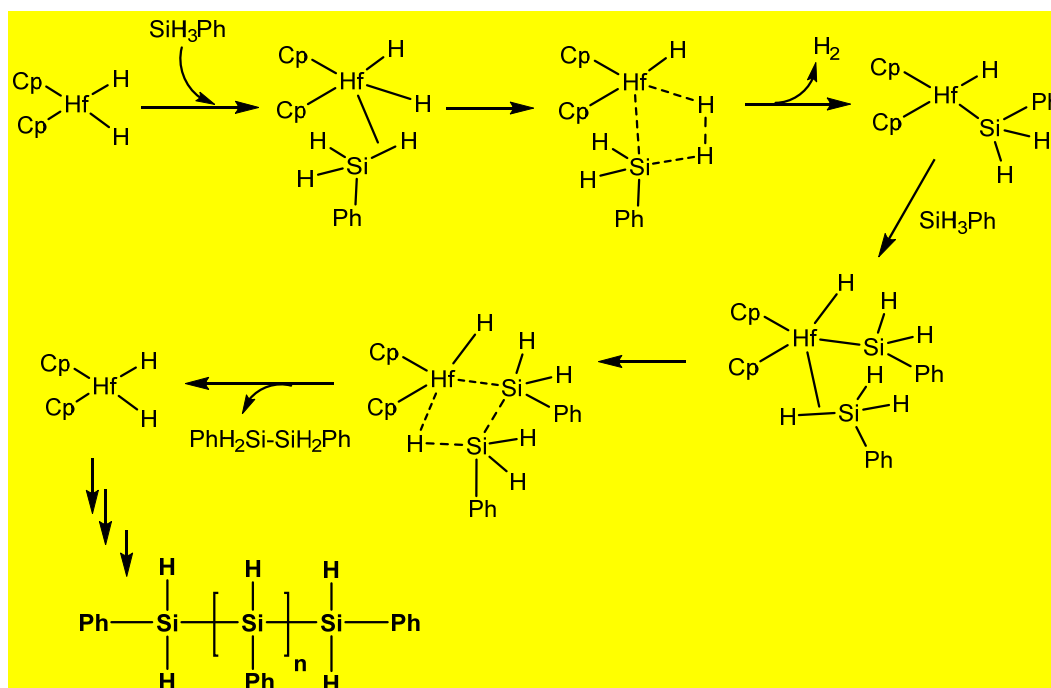
Section 5. Propose a reasonable mechanism for the following catalytic reactions (72 points).



- (1) Styrene coordination, 2;
- (2) Insertion, 3;
- (3) ethylene coordination 2;
- (4) Insertion, 3;
- (5) β -H elimination, 3
- (6) Release product 2

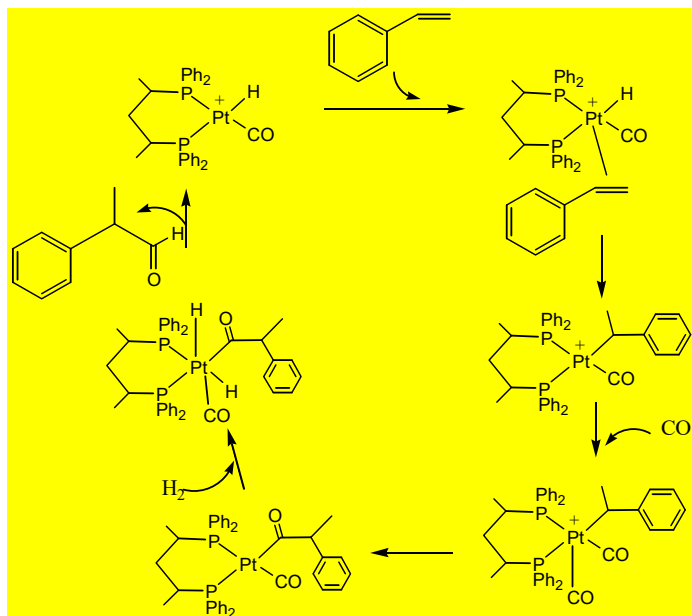
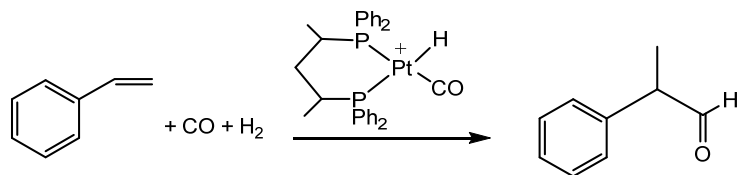


Or

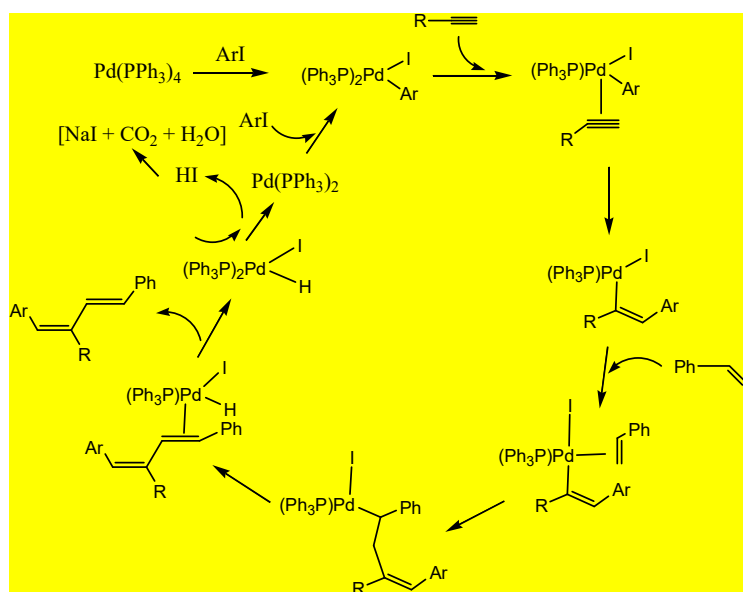
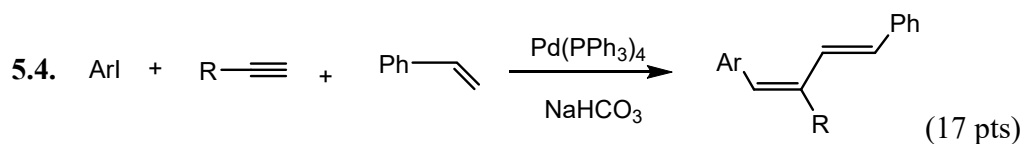


如果写成 Cp_2HfH_2 直接氧化加成，再还原消除的扣分

5.3 (Reference: *J. Am. Chem. Soc.* **2004**, 126, 5585) (15 pts)



- (1) Alkene Coordination 2;
- (2) H insertion, 3;
- (3) CO coordination, 2;
- (4) CO insertion, 3
- (8) H₂ OA, 3;
- (9) RE, 2



- (1) OA, 2;
- (2) π -alkyne 2;
- (3) Insertion, 2;
- (4) π -olefin 2;
- (5) Insertion, 2;
- (6) β -H elimination, 2
- (7) Release butyldiene 1
- (8) Elimination of HI, 2
- (9) NaHCO₃ base neutralization, 1
- (10) OA of ArI, 1